

Case Study

Project: LLM-Based Customer Support Chatbot (POC)

Organization: Aptos India

Role: Machine Learning Engineer

Problem

Support teams managing a B2B retail analytics dashboard were spending significant time handling repetitive, knowledge-heavy queries. Existing workflows had three critical issues:

- High manual dependency for common support questions
- Inconsistent answer quality due to scattered internal documentation
- Limited ability to scale high-quality responses across growing customer usage

Goal

Build an internal LLM-powered support assistant that can provide context-aware, reliable answers from internal sources while improving response quality and reducing manual effort.

System Design

Architecture components

- API layer for request orchestration and response delivery
- Retrieval pipeline over internal knowledge and analytics metadata
- Prompt templates with grounding constraints
- Evaluation and monitoring loop for output quality
- Containerized deployment for controlled experimentation and scale

Core technical choices

- Python + FastAPI for service orchestration
- RAG workflow using vector retrieval and context selection
- LangChain/LangGraph-based orchestration patterns
- Vertex AI and managed cloud components for operational reliability
- Dockerized deployment integrated with MLOps practices

Execution Approach

1. Retrieval quality first I started by improving retrieval relevance so prompts received precise context. This reduced unsupported generations and improved factual grounding.

2. Prompt versioning and evaluation I introduced structured prompt versions and compared them through offline tests and runtime metrics to improve consistency and tone while preserving factual correctness.

3. Production readiness I containerized the pipeline and added monitoring for failure patterns, response quality drift, and operational stability to support iterative improvement.

Results

- Reduced incorrect answers by 35% through retrieval and prompt optimization
- Reduced dependency on manual support teams by 30%
- Enabled faster iteration and more stable deployment via containerized MLOps workflows

What This Demonstrates for AISEO

This project demonstrates practical capability in building LLM-powered content/answer systems where quality, relevance, and operational reliability are equally important. The same approach translates well to GEO/SEO tooling: retrieval-driven answer quality, prompt-controlled output behavior, measurable evaluation loops, and scalable backend architecture.

Technology Stack

Python, FastAPI, RAG, LangChain, LangGraph, Vertex AI, Vector DB, Docker, MLOps.